Two-time postulate (t_+ for +U, t_- for −U): exploration of the five master equations

Compiled 2026-10-20, about 18:50 CDT. **Everything here is Category B**: exploration within the framework, with no derivation claimed. Items marked **[checked]** are algebra or numerics verified by the scripts in this folder; they check the *math of the exploration*, not physics. Items marked **[interp]** are interpretation. RNT (539 / 18+521 / 4880) is kept separate and nothing below uses it. No TPC or RH claims. Scripts: `two_time.py` → `two_time_output.json`; `bauer_map.py` → `bauer_map_output.txt`.

0. Sources

- Master equations, verbatim: `QC/grok-chats/raw-neutrino/043a2e2e…txt` L179 (E_leak), L182 (E_cosmos), L185 (F_friction), L188 (μ). Also `c39b5dc0…txt` L16-31, L333-347. V_flavor: `grok-chats/extracts/bounce/src-17-theorems-v5.txt` L210 and `3110c8b2…txt` L307.
- Chain: `QC/drive/extracts/theorem-lemma-chain-9-maths.md`.
- Two-time QM: M. Ridley and E. Cohen, "Two times or none?", arXiv:2509.22264 (Phil. Phys. 2026, doi:10.31389/pop.306). Fetched from arxiv.org/pdf/2509.22264. Equation numbers are from the arXiv v1 PDF.
- **Audit flags not built on**: A1 (TTC $T^2 \neq 0$) and A6 (539.9 vs 540 vs 539). The 540 appearing in E_cosmos, E_leak and F_friction is treated **as written but flagged**. Any result that depends on 540 ≠ 539.9 is conditional on A6 being intentional rather than a typo.

1. Original and two-time forms

Notation: $\omega_G = 2\pi/539.9\ s^{-1}$. "Reversed" means the −U sector runs in $-t_-$.

Eq	Original (single t)	Separable (S)	D2-coupled (C)	Locked (L): $t_- = -t_+ + \varphi$
E_leak	$E_0 e^{-t/\tau} L\Omega_{11} + \kappa_{dark}\sqrt{\delta(t)} v_0 f_{energy}(1+\rho_{DM}/10)(1+\Sigma \text{couplings}(t)) + \rho_{DM}\int_0^{t/(1+g_{11}\Phi)} \sin(\ldots)d\tau + \Sigma_z E_b(E_{z^-} - Z_{z^-})/(c^2\tau^-)\cdot\sin(2\pi t/(540+\Delta t_{11})) + \ldots$ (L179)	+U terms in t_+; the −U isotope sum in t_-: $\sin(-2\pi t_-/(540+\Delta t_{11}))$	S plus a transfer term $\kappa_{dark}\cdot X_+(t_+)X_-(t_-)$ evaluated on the brane	Every term becomes a function of t_+. The −U sum becomes $-\sin(2\pi t_+/(540+\Delta t) - \varphi')$
E_cosmos	$N_b\varepsilon(\Sigma_j[-m_j c^2\tau_-(t)e^{-t/\tau_-}\cos(2\pi t/(540h_j(1\pm\delta(t)/540))k_{seg})] - U)k_{bio} k_{mat} + E_{ZPE}\eta$ (L182)	The bracket is −U content, so it moves to t_- (cos is even, so reversal only flips the damping e^{+t_-/τ_-}). $E_{ZPE} \propto \sin(\omega_G t_+)$	S plus $N_b \varepsilon g X_+X_-$ (the brane count weights the transfer)	One time. The 540 term beats against 539.9 (§2.1)
F_friction	$0.9(\gamma-1)m\, v_0^2(1000/540)e^{-t/(\tau+\gamma N)}L + \Sigma_z \beta_z(N_{z^-} - 2Z_{z^-})e^{-t/\tau^-} + \ldots + \delta a_\mu^{-U}(\ldots)e^{-t/\tau^-}$ (L185; 1000/539 in c39b5dc0 L340, cf. A6)	+U damping in t_+; −U β-sum and δa_μ^{-U} in t_-	Friction = the transfer term itself (N-SFA: $F_{friction} \sim J_-(t)\delta a_\mu^{-U}$)	One time
V_flavor	$\Sigma_q m_q g_s X^M(t) \sin(\omega_G t) e^{-t/\tau}(1+\eta \text{ mod})$	Purely +U, so t_+ only	Unchanged unless X^M is the brane mode	Unchanged
μ	$1 + \kappa_{dark} f_{mean}(1+\rho_{DM}/10)/E_0\ \Omega_{11}(\ldots) + \ldots$ (L188). **No explicit t** (a mean)	Mean over the (t_+,t_-) torus. Equal to the single-t mean when the frequencies are incommensurate (ergodic)	Same, plus a g^2 correction	Same as the original

[interp] The −U-labelled terms already sit visibly inside the single-t equations: the mirror isotopes ($N_{z^-} - 2Z_{z^-}$), τ_-, δa_μ^{-U} and J_-. The two-time postulate amounts to giving those terms their own argument.

2. What the second time adds

2.1 Beats and torus orbits [checked]

- Drivers at 539.9 s (t_+) and 540 s (t_-, E_cosmos with $\delta = 0$) have ratio 5399/5400, so the (t_+,t_-) orbit is closed. On any one-time slice the beat and closure period is **2,915,460 s = 33.74 d**. *Conditional on A6*: if 540 is a typo for 539.9, there is no beat.

- With $\delta(t) \neq 0$, or Δt_{11} irrational, the orbit is quasi-periodic and dense on T^2.
- Separable and unlocked: observables are functions on T^2. The single-t TTC Lemma 3.1, $O(t+G_4) = O(t)$, holds **separately** in t_+ and in t_-, but not along a diagonal unless the periods are commensurate.

2.2 Lock reduces to one time [checked]

$\sin(2\pi(-t+\varphi)/T) \equiv -\sin(2\pi t/T - 2\pi\varphi/T)$, verified symbolically. Under (L), the second time is a **pure phase and sign**: no new dynamics, only a $-U$ phase offset φ.

2.3 Coupled sectors, conserved energy and stability [checked]

Toy model consistent with N-SFA's negative sign for $-U$: $L = \tfrac{1}{2}(\dot{a}^2 - \omega_1^2 a^2) - \tfrac{1}{2}(\dot{b}^2 - \omega_2^2 b^2) - g \cdot ab$, with $\omega_1 = 2\pi/539.9$ and $\omega_2 = 2\pi/540$.

- The conserved quantity is $E = E_+ - E_- + gab$. $dE/dt = 0$ on shell (symbolic), and numerical drift is about 3×10^{-12}.
- The characteristic polynomial is $\Omega^4 - (\omega_1^2 + \omega_2^2)\Omega^2 + \omega_1^2\omega_2^2 + g^2 = 0$, so the frequencies become complex when $\mathbf{g > |\omega_1^2 - \omega_2^2|/2 = 2.51\times10^{-8}\ s^{-2}}$ (only $1.85\times10^{-4}\ \omega_G^2$). Numerically, at $g = 0.5\ g_c$ the amplitude stays bounded (max $|a| = 1.44$). At $g = 2\ g_c$ both sectors grow together (max $|a| = 31.9$ over 2×10^6 s) while E stays conserved.
- [interp] This is the standard ghost (Pais–Uhlenbeck-type) behaviour: positive- and negative-energy sectors can exchange energy without bound. Because 539.9 and 540 are nearly degenerate, almost any brane coupling sits above threshold. If the two periods are equal (A6 typo), $g_c = 0$ and *every* coupling is unstable. **This is the main strain of the coupled variant.**

2.4 Unitarity and well-posedness [interp]

- Fields depending on both times obey ultrahyperbolic equations $(\partial^2_{t_+} + \partial^2_{t_-} - \nabla^2)\Phi$, which are generally ill-posed as initial-value problems unless constrained (Craig & Weinstein, Proc. R. Soc. A 465, 3023 (2009)).
- Unitarity holds only for the variants that reduce to a single evolution parameter (L, or the Bauer map in §4).

3. Consistency with the 9 Maths chain

Link	Fit
Axiom 0.1: $G_4 = 539.90$ s	One period. Two times need either a shared G_4 (fits) or a second period, which is **not established** (only via 540, A6)
TTC L1 / Lemma 3.1	Fits per time. A diagonal (joint) periodicity is **not established**. The TTC differential itself carries flag A1
N-SFA $H_+ \oplus H_-$, $\langle\cdot,\cdot\rangle_- = -\langle\cdot,\cdot\rangle_+$, $J_- = -e^{4\pi it/539.9}$	Fits the sector split. A negative norm combined with coupling **strains** (ghost, §2.3). J_- runs at the 2nd harmonic of a *single* t, so in the chain the $-U$ current is written in the shared time
B-MMT $d\mu \propto e^{2\pi it/539.9}$ $dx^4 \wedge \mathbf{d^3t} \wedge d^2\sigma$	The chain writes **three** time differentials, not two. A one-phase factor is used for the brane. Two-time **strains** this; the meaning of d^3t is **not established**
F-C PDE: unique periodic attractor $O(t) = a\cos + b\sin + c$	(S): attractor becomes T^2. (L): unchanged. (C): the attractor survives only below g_c. "Unique" **strains** for (S) and (C)
RTTC harmonics {5,…,45}, {1080,…,5400} s	Fit in each time. Cross-harmonics $m/539.9 + n/540$ are new and **not in the chain**

4. Comparison with Ridley & Cohen (Bauer two-time QM)

4.1 The paper's formalism (Section III, Bauer's approach)

- **State space**: the "Bauer space" $H_B = Sp\{|E\rangle\}_{E\geq0} \oplus Sp\{|E\rangle\}_{E\leq0}$ (Eq. 18), the negative-energy extension.
- **Energy shift**: $D(\varepsilon)$ (Eq. 19), a unitary group (Eqs. 21-26). Stone's theorem gives $D(\varepsilon) = e^{-i\varepsilon f}$ (Eq. 27), with $[f, H] = i$ (Eq. 29). Here $\tilde{H} = HQ = \mathrm{diag}(H, -H)$ (Eq. 16) and f has delta-normalised eigenstates (Eqs. 30-31, 38). The time operator splits into a forward part f_f (Eq. 35) and a backward part f_b (Eq. 36).
- **Pseudospin extension**: $H' = \mathrm{diag}(H, H) = H\mathcal{Q}$ (Eq. 41), $[f, H'] = iQ$ (Eq. 42). The spectrum is twofold degenerate, $|E, \alpha\rangle$ with $\alpha = f, b$ (Eqs. 43-44).
- **Evolution**: Heisenberg form (Eq. 45) gives $d\langle f\rangle/dt = \langle Q\rangle$ (Eq. 46). With projectors P (Eq. 47) and $P^{\perp} = I - P$ (Eq. 49): $df_f = dt$ (Eq. 48) and $df_b = -dt$ (Eq. 50).

- **Constraint and reduction**: there is **one** external parameter t. The two time degrees of freedom are slaved to it with opposite signs. The FPF/Keldysh "pinched present" is t = lim_{η→0} ½(t_f(η) + t_b(η)) (Eq. 51). The completeness Σ_α P_α = I is Eq. 52. Their contrast is the Page–Wootters constraint H_T|Ψ⟩⟩ = 0 (Eq. 2). The paper's final proposal combines timeless PW with two-time QM.
- Secondary sources. Kumar–Marto (CQG 2026, used in `grok-chats/raw/chat-01.txt` L432-447): one quantum state with forward and backward time components, with no extra coordinate. Bars 2T physics (hep-th/0008164): a (d,2) signature with a gauged Sp(2,R) that removes one time and one space dimension, leaving (d−1,1) "shadows".

4.2 Mapping: +U ↔ f, −U ↔ b, t_+ ↔ t_f, t_- ↔ t_b [interp, with checked algebra in bauer_map.py]

(L) Locked.

(i) This is exactly Eqs. 48/50: $dt_+ = dt$ and $dt_- = -dt$, so $t_- = -t_+ + \varphi$, where φ is the integration constant (or the Keldysh pinch offset of Eq. 51). The master equations become ⟨O⟩(t) = ⟨Ψ|P O_{+U} P + P O_{−U} P|Ψ⟩. The −U drivers enter as sin(ω_G·(−t) + φ′), so §2.2 holds.

(ii) *Matches*: two time degrees of freedom, one observable time, a unitary group, and a backward arrow for −U. *Differs*: Bauer's b-sector has negative energy with a **positive** norm (Eq. 18/16). N-SFA instead posits a **negative norm**. Bauer's version is unitary and ghost-free.

(iii) The paper adds a Hermitian time operator (Eq. 33) and fixes the locking as a theorem of the formalism, not a choice. **Verdict: congruent** (with N-SFA's norm re-read as Bauer's energy sign).

(S) Separable.

(i) $H' = \mathrm{diag}(H_+, H_-)$ keeps [Q, H′] = 0, so the sectors evolve independently, but still in **one** t. A genuinely independent second parameter t_- is not in the paper. Getting one would need a separate PW-type constraint per sector (Eq. 2 applied twice), which the paper does not do.

(ii) *Matches*: the block structure and the separate f/b components. *Differs*: the paper has no independent t_- and no T^2 torus. The beats of §2.1 would come only from $H_+ \neq H_-$ spectra (539.9 vs 540) inside one t, not from a second time.

(iii) **Partially congruent.**

(C) D2-coupled.

(i) Add brane transfer as an off-diagonal term: $H' + g\sigma_x$.

- [checked] Starting in f gives ⟨Q⟩(t) = cos 2gt. The eigenvalues are E ± g and the evolution is unitary.
- By Eq. 46, d⟨f̂⟩/dt = cos 2gt: the **arrow of time oscillates** between +U-forward and −U-backward.
- If $g = \pi/539.9\ s^{-1} \approx 5.82\times10^{-3}\ s^{-1}$, the arrow-mixing period equals G_4. That is a possible framework-level reading of "transfer modulated at 539.9 s".
- In the H̃frame (Eq. 16), the same coupling gives $E' = \pm\sqrt{(E^2 + g^2)}$ [checked]. That is also bounded and unitary.

(ii) *Differs from §2.3*: in Bauer's positive-norm version there is **no runaway**. The classical instability in §2.3 comes from the negative-norm / negative-kinetic Lagrangian. *Also differs from the paper*: the paper never couples f and b. Q is conserved there, and the coupling breaks Eq. 46's constancy of ⟨Q⟩.

(iii) **Partially congruent.** The formalism accommodates the coupling unitarily, but at the cost of Q-conservation. The paper itself does not propose this.

Secondary sources. Kumar–Marto is closest to (L) (two components, one time). Bars 2T is **incongruent** with all three variants: there is no (·,2) metric and no Sp(2,R) gauge, and (L) only loosely resembles a gauge fixing.

4.3 Comparison table

Variant	Paper's math used	Matches	Differs	Paper adds	Verdict (Cat B)
L locked	Eqs. 41-50, 51	2 time dof, 1 observable t, −U backward	N-SFA negative norm vs Bauer negative energy with positive norm	Time operator (Eq. 33); lock forced, not chosen; unitarity	**Congruent**

S separable	Eqs. 41-44	f/b block structure	No independent $t_⌴$, no torus	Would need a PW constraint (Eq. 2) per sector	**Partially congruent**
C coupled	Eqs. 16, 41, $46 + g\sigma_x$	Unitary transfer; can be tuned to a G_4 period	Q no longer conserved; coupling absent from the paper	Bounded spectrum, unlike the ghost classical model	**Partially congruent**
Bars 2T (ref.)	—	—	No Sp(2,R), no (d,2)	—	Incongruent

5. Testable or distinguishing consequences (Cat B, honestly marked)

1. **33.7-day beat** (2,915,460 s) between the 539.9 s and 540 s components in long time series that already show a G_4 line. This rests entirely on the framework's prior claim of a 539.9 s signal and on A6. It is not a prediction independent of those assumptions. The R-FFT engine in `QC/datasets` could look for sidebands at 1/539.9 ± 1/2,915,460 Hz.

2. **Locked variant**: no new observable. It differs from one time only by the phase φ of −U-labelled terms. That is not distinguishable unless −U terms can be isolated.

3. **Coupled (Bauer) variant**: a periodic reversal of the effective arrow, $d\langle f\rangle/dt = \cos 2gt$. No measurable handle on $\langle f\rangle$ is established.

4. **Coupled (N-SFA ghost) variant**: predicts runaway growth above $g_c \approx 2.5\times10^{-8}\ s^{-2}$. Its absence would argue for the Bauer-style positive-norm reading.

5. Not established: any link from $t_⌴$ to galactic structures, halos or PBH chirality.

6. Bottom line (Cat B)

- The second time adds real structure only if it is independent (S) or coupled (C).

- (S) yields a torus, with a 33.7 d beat that depends on A6.

- (C) yields a classical ghost instability that is easy to trigger near-degenerately, unless it is re-read in Bauer's positive-norm frame, where it becomes a unitary arrow oscillation.

- (L) collapses to one observable time with a −U phase. It is exactly the structure of Ridley & Cohen's Bauer two-time QM (Eqs. 46-51), and the most congruent with both the paper and the chain's single-phase B-MMT measure.
- Open: the meaning of B-MMT's d^3t, and whether N-SFA's negative norm should be read as Bauer's negative energy.